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CS355TBE - Quantum Computing

Introduction

Course In charge
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Quantum Computing

- Cutting-edge field of computer science
- **Quantum mechanics -** to process information
- Solve problems **exponentially faster** than classical computers.

Ideal for:

  - Cryptography (breaking or creating secure codes)
  - Drug discovery and molecular simulation
  - Optimization problems (logistics, finance, AI training)
  - Material science

- Companies like **IBM, Google, Rigetti, IonQ, Xanadu** are racing toward building large-scale fault-tolerant quantum computers.

Why Quantum Computing

- Quantum computing is required to solve **problems too complex, too time-consuming, or simply impossible for classical computers**—unlocking breakthroughs in science, security, and technology.

Optimization Problems

- Examples: airline scheduling, supply chains, financial portfolio optimization, traffic management.
- Classical methods **may take years**; quantum approaches can find **better solutions much faster**.

Artificial Intelligence & Big Data

- Machine learning models require **huge amounts of computation**.
- Quantum computers can accelerate training and pattern recognition, especially in **high-dimensional data.**

Classical vs Quantum Computing

Feature	Classical Computing	Quantum Computing
Basic Unit	**Bit** – can be 0 or 1	**Qubit** – can be 0, 1, or a superposition of both
Processing	Sequential (step by step)	Parallelism (can explore **many possibilities at once** due to superposition)
Information Storage	Deterministic: one definite state at a time	Probabilistic: multiple states until measured
Key Principle	Based on classical physics (Boolean logic, transistors)	Based on **quantum mechanics** (superposition, entanglement, interference)
Data Representation	Binary states stored in transistors (on/off)	Quantum states represented using **vectors in Hilbert space**
Operations	Logic gates (AND, OR, NOT, XOR)	**Quantum gates** (Hadamard, Pauli, CNOT, Toffoli, etc.)

Error Handling	Errors managed with classical error correction	Very sensitive to noise, requires complex **quantum error correction**
Speed/Power	Efficient for everyday tasks (word processing, browsing, databases)	**Powerful for specialized tasks** (cryptography, molecular simulation, optimization)
Examples	Laptops, smartphones, supercomputers	IBM Quantum, Google Sycamore, IonQ, Xanadu
Stage of Development	Mature, widely used worldwide	Emerging in research & early applications (NISQ era)

In general,

- A **classical computer** is like **reading a book line by line**.
- A **quantum computer** is like reading **all possible pages at once**

Applications of Quantum Computing

1. Cryptography & Cybersecurity

- **Breaking classical encryption:** Shor's algorithm can factor large numbers, threatening RSA and ECC security.
- **Quantum key distribution (QKD):** Provides unbreakable communication channels (used in quantum networks).
- **Post-quantum cryptography:** Developing new encryption methods resistant to quantum attacks.

2. Algorithms & Complexity

- **Quantum search algorithms:** Grover's algorithm speeds up searching in large databases.
- **Optimization algorithms:** Faster solutions for scheduling, routing, and resource allocation.
- **Machine learning & AI:** Quantum ML models for faster training and better handling of big data.

Applications of Quantum Computing

3. Software Engineering & Simulation

- **Quantum simulators:** Simulating quantum systems directly
- **Programming frameworks:** Qiskit (IBM), Cirq (Google), and Quipper are quantum programming languages.
- **Verification & testing:** New methods for debugging and testing quantum software.

4. Engineering & Applied Sciences

- **Electrical & Electronics Engineering:**
  - Quantum sensors for precise measurements.
  - Quantum communication systems (satellite-based secure communication).
- **Mechanical & Civil Engineering:**
  - Optimization of structural designs.
  - Simulation of materials with quantum-level accuracy.
- **Chemical & Materials Engineering:**
  - Discovering new materials, catalysts, and superconductors.
  - Drug discovery through molecular simulation.
- **Computer Engineering:**
  - Development of **specialized quantum processors.**
  - Quantum error correction hardware.

Applications of Quantum Computing

5. Emerging Applications

- **Artificial Intelligence & Data Science:** Quantum-enhanced neural networks and clustering.
- **Networking:** Quantum internet (secure communication).
- **Energy:** Simulation of nuclear fusion and efficient energy materials.
- **Healthcare:** Protein folding simulations for **drug/vaccine development.**

- Quantum computing in computer science and engineering enables **faster algorithms, stronger security, smarter AI, accurate simulations, and optimized engineering solutions**.

Course Learning Objectives (CLOs)

- Understand quantum mechanics foundations.
- Learn how qubits & gates work.
- Explore algorithms and cryptography.
- Get exposure to simulators/tools.
- Develop teamwork & communication skills in quantum research.

Unit 1

Unit-I	09 Hrs
Quantum Building Blocks: Quantum mechanics of Photon Polarization, Single Quantum bits, Single Qubit Measurement, A Quantum key Distribution Protocol, State Space of a Single-Qubit System, Direct Sums and Tensor Products of Vector Spaces, State Space of an n-Qubit System, Entangled States, MultiQubit Measurement, QKD using Entangled states.	

Unit 2

Unit – II	09 Hrs
Multiple-Qubit States Measurements: Dirac Bra/Ket Notation for Linear Transformation, Projection Operators for Measurement, Hermitian Operator Formalism for Measurement, EPR Paradox and Bell's Theorem.	

Unit 3

Unit –III	09 Hrs
Quantum State Transformations: Unitary transformations, No-Cloning Principle, Some Simple Quantum Gates, Pauli transformations, Hadamard Transformations, Multiple-Qubit Transformations, ControlledNOT and other singly controlled gates, Applications of Simple Gates, Dense coding, Quantum teleportation	

Unit 4

Unit –IV	09 Hrs
Introduction To Quantum Algorithms: Computing with Superpositions, Walsh-Hadamard transformation, Quantum Parallelism, Notions of Complexity, Query Complexity, Communication Complexity, Simple Quantum Algorithm- Deutsch's Problem.	

Unit 5

Unit –V	09 Hrs
Simple Quantum Algorithms: Deutsch-Jozsa Problem, Bernstein-Vazirani Problem, Simon's Problem, Machine Models and Complexity Classes, Shor's factoring Algorithm, Example illustrating Shor's Algorithm	

Course Outcomes

Course Outcomes: After completing the course, the students will be able to: -	
CO 1	Explain the various essentials of quantum computation, Qubits, and Quantum operators.
CO 2	Analyze the working of quantum transformations and quantum gates.
CO 3	Describe the principle of working of some of the quantum algorithms and conduct simulations using open-source quantum simulators.
CO 4	Investigate the applications of quantum computing algorithms and quantum cryptography in realworld applications.
CO 5	Appraise the knowledge and potential in quantum computing to build a successful career, work in teams, and communicate their ideas effectively.

Reference Books

Reference Books	
1.	Quantum Computing: A Gentle Introduction, Eleanor Rieffel and Wolfgang Polak, 2011, The MIT Press, ISBN 9780262015066.
2.	An introduction to Quantum Computing, Phillip Kaye, Raymond Laflamme, Muchele Mosca, Oxford University Press, 2007, ISBN-13: 978-0198570493, ISBN-10: 019857049X
3.	Quantum Computing for Computer Scientists, 1st Edition, Noson S. Yanofsky and Mirco A. Mannucci, Cambridge University Press, 2008, ISBN 978-0-521-879965.

The Future

- Tech giants (IBM, Google, Microsoft) and start-ups are racing to build scalable quantum computers.
- Governments and universities are **investing in research**.
- It could **transform industries** like cybersecurity, healthcare, finance, and artificial intelligence.
- By 2030, quantum computing is projected to create **millions of jobs globally**.
- Demand will be high in **AI, cybersecurity, finance, healthcare, materials, and telecom**.

Course Learning and Expectations

- Evaluation through assignments, case studies, quiz, exams and EL.
- Encouragement for **hands-on practice** in virtual environments.
- **Industry collaboration with Unisys** – Learn from experts and access real-world case studies, webinars, industry visit, 2 days' workshop, Guidance and support for EL, Project work, Expert lectures, partial delivery from Quantum scientists.
- **Early advantage in an emerging field** – Be among the first to explore Quantum Computing, a technology expected to revolutionize multiple industries.
- **Career-ready skills** – Strengthen your profile with knowledge that leading companies and research labs are already investing in.
- **Hands-on and applied learning** – The course design blends academic foundations with industry perspectives to make the subject accessible and impactful.
- **Future-proof opportunities** – Quantum Computing is **opening doors in many fields but not limited to** cryptography, AI, optimization, pharmaceuticals, finance, and beyond.

Thank you